

BIRMINGHAM, United KingdomWMO: **035340**Lat: **52.454N**Lon: **1.748W**Elev: **327**StdP: **14.52**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.4	26.7	20.6	15.6	28.2	22.8	17.4	30.4	26.7	47.2	23.8	46.4	3.8	310	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	80.2	64.6	75.8	62.9	73.1	61.8	66.4	75.9	64.7	73.0	63.2	70.5	8.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	86.8	69.2	61.2	81.7	67.3	60.5	79.7	66.5	31.2	76.0	29.8	72.8	28.8	70.1	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.4	19.0	17.1	DB	17.9	85.7	4.6	4.0	14.6	88.6	11.9	90.9	9.4	93.2	6.1	96.1	(4)
(5)			WB	17.4	69.8	4.7	2.1	14.0	71.3	11.3	72.5	8.7	73.7	5.3	75.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	39.9	40.5	43.4	47.4	52.9	58.5	62.6	61.9	57.2	51.4	44.7	40.5	(6)	
	DBStd	9.74	5.97	5.90	5.49	5.29	5.12	4.68	4.58	4.36	4.25	5.25	5.64	6.92	(7)	
	HDD50	1452	315	268	213	109	28	1	0	0	2	46	172	298	(8)	
	HDD65	5486	779	685	670	528	375	202	104	117	237	423	608	758	(9)	
	CDD50	1503	1	3	8	31	118	257	391	369	218	88	14	5	(10)	
	CDD65	62	0	0	0	0	0	8	30	22	2	0	0	0	(11)	
	CDH74	554	0	0	0	1	12	90	272	158	18	3	0	0	(12)	
	CDH80	97	0	0	0	0	0	13	50	33	1	0	0	0	(13)	
(14) Wind	WSAvg	8.7	9.5	9.6	9.4	8.8	8.7	8.3	8.0	7.8	8.0	8.6	8.5	8.8	(14)	
(15) Precipitation	PrecAvg	26.5	2.5	1.9	1.9	1.9	2.1	2.2	2.0	2.6	2.3	2.1	2.4	2.6	(15)	
	PrecMax	35.9	5.2	5.5	3.5	4.1	6.7	5.7	4.3	4.7	5.4	4.9	5.2	5.4	(16)	
	PrecMin	19.4	0.6	0.3	0.2	0.1	0.5	0.4	0.2	0.4	0.4	0.2	1.0	0.5	(17)	
	PrecStd	3.6	1.1	1.1	0.9	1.0	1.3	1.3	1.1	1.1	1.4	1.2	1.0	1.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	57.0	61.2	70.0	75.5	82.1	85.6	84.2	76.7	67.9	59.3	56.9	(19)	
		MCWB	52.4	51.9	52.1	57.1	61.9	65.3	65.9	65.3	64.1	60.8	56.2	53.6	(20)	
	2%	DB	53.0	53.6	57.3	64.3	71.4	76.5	80.5	78.6	71.5	63.5	57.3	54.2	(21)	
		MCWB	50.4	50.1	50.5	54.3	59.9	62.6	64.5	64.6	61.9	58.3	54.6	51.8	(22)	
	5%	DB	50.3	51.6	55.1	60.6	66.6	71.8	76.9	73.7	68.1	61.1	55.4	52.3	(23)	
		MCWB	48.2	48.2	49.7	52.4	57.4	60.9	63.3	62.7	60.1	57.0	52.6	50.0	(24)	
	10%	DB	48.5	49.7	52.2	57.2	62.9	68.2	73.2	70.2	65.8	59.2	53.4	50.3	(25)	
		MCWB	46.7	47.0	47.6	50.3	55.0	59.1	61.8	61.2	58.3	55.6	50.9	48.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	53.0	54.4	58.3	63.8	68.2	68.4	68.0	66.0	62.1	57.3	54.4	(27)	
		MCDB	55.1	55.7	58.2	67.0	71.6	77.4	79.4	77.4	73.5	66.6	58.9	56.2	(28)	
	2%	WB	50.9	50.7	52.2	55.3	61.0	64.3	66.3	65.7	63.5	59.8	55.2	52.8	(29)	
		MCDB	52.6	52.8	55.7	62.1	68.7	72.1	76.8	74.4	69.4	63.0	57.2	54.7	(30)	
	5%	WB	48.8	48.9	50.4	53.1	58.8	62.1	64.6	64.2	61.5	58.0	53.3	51.0	(31)	
		MCDB	50.1	50.9	53.9	59.3	65.6	69.5	73.7	71.6	66.2	60.7	55.1	52.7	(32)	
	10%	WB	46.8	46.9	48.6	51.2	56.5	60.2	62.9	62.8	59.7	56.3	51.4	48.7	(33)	
		MCDB	48.3	49.0	52.1	56.3	62.4	66.9	71.3	69.0	64.0	59.0	53.2	50.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	10.7	13.2	15.9	16.0	16.0	16.6	15.9	14.9	12.4	10.3	9.3	(35)	
		MCDBR	10.9	11.6	16.7	21.6	22.5	22.6	23.4	21.5	19.4	13.6	10.6	9.9	(36)	
	5% WB	MCWBR	10.3	9.4	11.3	13.2	13.2	11.7	10.2	9.9	11.0	9.6	9.3	9.5	(37)	
		MCWBR	10.5	10.7	14.8	19.1	20.1	19.0	20.1	17.8	16.5	12.4	10.3	9.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.321	0.337	0.368	0.394	0.399	0.396	0.395	0.381	0.366	0.348	0.323	0.310	(40)	
		taud	2.362	2.391	2.299	2.237	2.261	2.294	2.301	2.363	2.408	2.436	2.413	2.338	(41)	
	Ebn at Noon		212	246	259	266	270	272	270	267	258	238	210	192	(42)	
		Edh at Noon	19	25	33	39	40	39	38	34	29	23	18	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		238	455	798	1224	1512	1618	1544	1272	956	552	291	192	(44)	
	RadStd		21	54	96	135	95	163	168	93	83	48	32	20	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (95 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page